Debamitro Chakraborti

CONTACT

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NOW

DerrickCPP
Founder
Building the world's best automated C++ code review product for enterprises

East Coast Software LLC
Owner
Building cross-platform desktop software. Providing basic AI services and training to small businesses and individuals

PRODUCTS

HTML Drive
A web app for publishing HTML files stored on Google Drive

CodeAgent
A tiny portable clone of Claude Code/Open Code, written in Rust

FindBigFolders
A cross-platform desktop app for free'ing up disk space, written in Rust

Durga Puja Map of USA
An iOS app providing a map of an Indian cultural festival in USA

PROFESSIONAL EXPERIENCE

Cadence Design Systems

Xcelium
This product simulates hardware by compiling gigabytes of high level source code. Even the most complex chips in the world use this product. It is best to describe this product as a compiler toolchain for a complex high-level language which is only used to create massive codebases. Before you ask, yes this is how GPUs are designed.

Software Architect **August 2024 - December 2024**

Paving the path for Xcelium runtime performance improvements for the next decade

Senior Principal Software Engineer **October 2020 - July 2024**

- Leading a team of engineers working on Xcelium performance
- Technical lead for a brand new optimization engine for Xcelium
- Working to speed up Xcelium for faster verification of next-generation hardware designs

Principal Software Engineer **August 2010 - September 2020**

- Optimized the runtime performance of the simulator by as much as 10x in various critical customer designs.
- Improved the peak memory usage of the compiler by as much as 30%.
- Wrote IR-based compiler optimizations including function inlining for the System Verilog language.
- Implemented new scheduling semantics in the simulator.
- Designed and developed various parts of the runtime library of the simulator.

- Refactored legacy code in a ~10MLOC codebase.
- Started introducing modern C++ in the team and organized technical learning sessions.
- Took initiative for improving the build system to make it faster by parallelization.

Nusym Technology India Pvt Ltd **R&D Engineer** **October 2008 - June 2010**

- Developed various parts of the interpreted simulator.

Atrenta India Pvt Ltd **Senior Software Engineer** **July 2005 - October 2008**

- Worked on the core engine of a static analysis tool.

Education and awards

- Recurse Center, New York City (Mini batch, August 2018)
- B.E. in Computer Science and Engineering (year: 2005) from Jadavpur University, Kolkata (percentage: 75)
- 5th rank in West Bengal Joint Entrace Examination (2001)
- Jagadis Bose National Science Talent Search scholar (batch of 2001)